

Question_10:

Code:

```
1  /*
2  Question_10:
3  Check whether a line parallel to  $y = x$  can divide
4  the people equally on both sides.
5  */
6
7  #include <iostream>
8  #include <vector>
9  #include <algorithm>
10 using namespace std;
11
12 int main() {
13     int t;
14     cin >> t;
15
16     while (t-- > 0) {
17         int n;
18         cin >> n;
19
20         vector<pair<int, int>> a(n);
21         int l = 10000, r = -10000;
22
23         for (int i = 0; i < n; i++) {
24             int x, y, h;
25             cin >> x >> y >> h;
26
27             a[i] = {y - x, h};
28
29             l = min(l, y - x);
30             r = max(r, y - x);
31         }
32         bool ok = false;
33         while (l <= r) {
34             int mid = (l + r) / 2;
35
36             long long left = 0, right = 0, same = 0;
37
38             for (int i = 0; i < n; i++) {
```

```
39         if (a[i].first < mid)
40             left += a[i].second;
41         else if (a[i].first == mid)
42             same += a[i].second;
43         else
44             right += a[i].second;
45     }
46     if (left == right) {
47         ok = true;
48         break;
49     }
50     left += same;
51     if (left == right) {
52         ok = true;
53         break;
54     }
55     if (left > right)
56         r = mid - 1;
57     else
58         l = mid + 1;
59 }
60 cout << (ok ? "YES" : "NO") << endl;
61 }
62 return 0;
63 }
```

Output:

Test Case_1:

```
3
3
-2 1 1
-1 1 3
1 -1 4
YES
3
-2 1 1
-1 1 1
-2 1 1
NO
4
-2 1 1
-1 1 2
-2 1 1
3 1 5
NO
PS C:\Users\Prabin Yadav\
```

Test

Case_2:

```
3
3
1 -1 4
-1 1 3
1 -1 4
NO
3
1 -1 4
-1 1 1
1 -1 4
NO
4
1 -1 4
-1 1 2
1 -1 4
3 1 5
NO
PS C:\Users\Prabin Yadav\
```